# SOFTWARE REQUIREMENTS SPECIFICATION

Version 1.0

## ONLINE TEACHER EVALUATION SYSTEM

Cebu Mary Immaculate College (CMIC)

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*Submitted in partial fulfillment of the requirements for the Analysis Phase*

## Document Information

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# 1.0 Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) describes the proposed functions, interfaces, data, and quality requirements of the Online Teacher Evaluation System for Cebu Mary Immaculate College (CMIC). It serves as a shared reference for the project stakeholders and developers during analysis, design, implementation, and testing.

The document identifies what the system is expected to do, who will use it, the information it will handle, and the constraints under which it must operate. It is intended to reduce misunderstandings by stating the system requirements in a clear and organized manner.

## 1.2 Scope of Project

The Online Teacher Evaluation System is a web-based application intended to support the evaluation of teachers by students. It is designed to replace or improve parts of a manual paper-based process, which may involve delayed distribution, manual tallying, inconsistent timing, and concerns about students expressing honest feedback.

The system is intended to provide role-based access for administrators, students, and teachers. Administrators manage system and academic records and control evaluation periods. Students can access available evaluation periods, submit evaluations for assigned teachers, and review their own profile. Teachers can access their profile and view evaluation results made available to them. Evaluation responses should be handled confidentially, and reports should present results in a way that does not unnecessarily reveal the identity of a student.

The system is intended to make evaluation access more convenient, reduce manual computation, organize feedback, and help the institution use evaluation results to support teaching improvement. Exact institutional policies, evaluation questions, and access rules must be confirmed by CMIC before deployment.

## 1.3 Glossary

| Term | Definition |
|---|---|
| Administrator | Authorized user who manages accounts, academic records, evaluation periods, and system settings. |
| Student | User who logs in to complete teacher evaluations during an active period. |
| Teacher | User who can access their account and view evaluation results permitted by the institution. |
| Evaluation | A student's submitted responses about a teacher, based on the institution-approved evaluation form. |
| Evaluation period | A defined school year/semester window during which evaluations are available. |
| Evaluation form | The set of approved criteria or questions used to evaluate a teacher. |
| Evaluation result | A summary or report of submitted ratings and feedback. |
| Anonymous response | A response presented without exposing the student's identity to the teacher; implementation and access rules must follow institutional policy. |
| Database | Organized collection of system records. |

| | |
|---|---|
| Role-based access | Access permissions assigned according to whether the user is an administrator, student, or teacher. |
| SRS | Software Requirements Specification; a document describing a software system's requirements. |

## 1.4 References

IEEE Computer Society. (1998). IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

ISO/IEC. (2011). ISO/IEC 25010: Systems and software Quality Requirements and Evaluation (SQuaRE) – System and software quality models. The project may use relevant quality characteristics as evaluation criteria.

Project-specific information supplied for the Online Teacher Evaluation System, including the identified user roles, evaluation workflow, and planned system modules.

## 1.5 Overview of Document

Section 1 introduces the project, its purpose, scope, terminology, and references. Section 2 describes the system environment, principal use cases, user characteristics, and general quality expectations. Section 3 provides more detailed functional requirements, interface requirements, logical data descriptions, and non-functional requirements.

# 2.0 Overall Description

## 2.1 System Environment

The system is a web application accessed through a modern web browser over a network. It uses a server-side application and a relational database to authenticate users, manage academic and evaluation records, and display role-specific pages. Internet or local network availability, server configuration, and database availability affect access to the application.

| Actor / Component | Interaction with the System |
|---|---|
| Administrator | Signs in to manage user records, courses, departments, evaluation forms/periods, and system information. |
| Student | Signs in, views available evaluations, submits responses, and accesses permitted profile information. |
| Teacher | Signs in, accesses profile information, and views evaluation results that have been released or made available. |
| Web browser | Displays pages and sends user actions to the application. |
| Application server | Processes authentication, validation, business rules, and requests. |
| Relational database | Stores user accounts, academic information, evaluation periods, forms, submitted evaluations, and related details. |

System context (textual representation): Administrator, Student, and Teacher → Web Browser → Online Teacher Evaluation System → Relational Database.

## 2.2 Functional Requirements Specification

This section summarizes the primary use cases for each user role. Detailed requirement statements are provided in Section 3.2.

### 2.2.1 Administrator Use Cases

#### Use case: Sign In

Brief description: The administrator enters valid credentials to access the administration area.

Precondition: An administrator account exists and the login page is available.

Main steps:

1. The administrator opens the login page.
2. The administrator enters a username and password.
3. The system validates the credentials and account role.
4. If valid, the system opens the administrator dashboard; otherwise, it displays a general error message.

Postcondition: An authenticated administrator session is established, or access is denied.

#### Use case: Manage Users

Brief description: The administrator creates, views, updates, and deactivates user records for administrators, students, and teachers, subject to permissions.

Main steps: The administrator opens the appropriate user module, chooses an action, enters or edits required information, submits the form, and the system validates and saves the record. The system displays a success or validation message.

#### Use case: Manage Academic Information

Brief description: The administrator manages departments, courses, and related academic records used by the system.

Main steps: The administrator opens the relevant module, adds or selects a record, enters the required details, saves changes, and reviews the updated list.

#### Use case: Manage Evaluation Period

Brief description: The administrator defines the school year, semester, opening date, closing date, and status of an evaluation period.

Main steps: The administrator opens Evaluation Periods, creates or edits a period, checks the dates and status, and saves it. The system makes evaluations available only when the period is active and open according to configured rules.

#### Use case: Manage Evaluation Form

Brief description: The administrator maintains the approved evaluation questions and rating criteria used by students.

Main steps: The administrator opens evaluation-form management, adds or edits questions and rating instructions, validates the form, and saves it for use in an evaluation period.

### 2.2.2 Student Use Cases

#### Use case: Submit Teacher Evaluation

Brief description: The student evaluates an eligible teacher during an active evaluation period.

Precondition: The student is authenticated, the period is open, and the student has an evaluation available to complete.

Main steps:

5. The student opens My Evaluations.
6. The system displays eligible teachers or pending evaluations.
7. The student selects a teacher and reads the evaluation instructions.
8. The student answers the required questions and optionally enters written feedback if enabled.
9. The student reviews and submits the responses.
10. The system validates the submission, stores the evaluation, and confirms completion.

Postcondition: The evaluation is recorded and the student cannot submit a duplicate evaluation for the same required teacher and period unless an authorized rule permits it.

#### Use case: Manage Student Profile

Brief description: The student views permitted profile details and updates editable fields, such as contact information or profile picture, if supported.

### 2.2.3 Teacher Use Cases

#### Use case: View Evaluation Results

Brief description: The teacher opens the evaluation results page to view released summaries for the relevant period.

Precondition: The teacher is authenticated and results are available to that account under institutional policy.

Main steps: The teacher opens Evaluation Results, selects an available period if applicable, and the system displays permitted summaries. The system must not expose student identity through results unless the institution has explicitly authorized and documented such access.

Postcondition: The teacher views only results authorized for their account.

#### Use case: Manage Teacher Profile

Brief description: The teacher views profile details and updates permitted fields.

## 2.3 User Characteristics

| User | Expected Characteristics |
|---|---|
| Administrator | Has basic computer and browser skills and is authorized to maintain system data. Should understand the institution's evaluation procedures. |
| Student | Has basic skills using a browser, can sign in, read instructions, and complete a digital form. |
| Teacher | Has basic skills using a browser and can navigate profile and results pages. |

## 2.4 Non-Functional Requirements

- The interface should use clear labels, consistent navigation, readable text, and understandable validation messages.
- The system should validate required fields and prevent incomplete or duplicate submissions where applicable.
- The system should restrict access according to user role and protect credentials and evaluation records.
- The system should provide reasonable response times under normal institutional usage; exact performance targets should be established during testing.
- The system should preserve valid records during ordinary use and display an appropriate message when an operation fails.
- The application should be maintainable so that authorized developers can update forms, rules, and modules without unnecessary changes to unrelated features.

# 3.0 Requirements Specification

## 3.1 External Interface Requirements

### 3.1.1 User Interface

The system shall provide a login page and role-specific dashboards for administrators, students, and teachers. Navigation items shall be relevant to the signed-in role. Forms shall identify required fields, provide appropriate input controls, and display validation or status messages. Destructive actions, such as deleting or deactivating a record, should require confirmation.

### 3.1.2 Hardware Interface

No specialized hardware is required beyond a computer, tablet, or smartphone capable of running a supported web browser and connecting to the application server.

### 3.1.3 Software Interface

The system shall interact with a relational database through the application's configured database connection. It shall run on a supported server environment with the required runtime and dependencies installed. Exact hosting, browser, and server versions must be documented before deployment.

### 3.1.4 Communication Interface

Users access the application through HTTP over the configured network. For production deployment, HTTPS should be used to protect credentials and submitted information in transit.

## 3.2 Functional Requirements

The requirements below are written as proposed system requirements. They should be confirmed with the project adviser and CMIC stakeholders before final approval.

### 3.2.1 Login and Logout

| Field | Specification |
|---|---|
| Requirement ID | 3.2.1 |
| Trigger | A user opens the login page or chooses to log out. |
| Precondition | The login page is available; a valid account is required for protected pages. |
| Basic Path | 1. The user enters the required credentials.\n2. The system validates the credentials and determines the user role.\n3. The system starts an authenticated session and redirects the user to the correct dashboard.\n4. When the user chooses logout, the system ends the session and returns to the login page. |
| Alternative Paths | If credentials are invalid, the system denies access and displays a clear message. |
| Postcondition | A valid user is signed in, or the session is ended after logout. |
| Exception Paths | The system must not reveal whether a particular username or password was correct. |
| Other | Protected pages must not remain accessible after logout without a new authenticated session. |

### 3.2.2 Manage Users

| Field | Specification |
|---|---|
| Requirement ID | 3.2.2 |
| Trigger | The administrator selects a user-management module. |
| Precondition | The administrator is authenticated and authorized. |
| Basic Path | 1. The system displays the list of records in the selected module.\n2. The administrator chooses to add, view, edit, or deactivate a record.\n3. The administrator enters the required details.\n4. The system validates required fields and uniqueness rules.\n5. The system saves valid changes and refreshes the displayed list. |
| Alternative Paths | If required information is missing or a unique field conflicts, the system identifies the issue and does not save invalid data. |
| Postcondition | The user record is added or updated, or deactivation is recorded. |
| Exception Paths | If the database operation fails, the system reports failure and preserves the previous valid record. |
| Other | Passwords must not be stored in plain text. |

### 3.2.3 Manage Academic Information

| Field | Specification |
|---|---|
| Requirement ID | 3.2.3 |
| Trigger | The administrator opens Departments or Courses. |
| Precondition | The administrator is authenticated and authorized. |
| Basic Path | 1. The system displays existing academic records.\n2. The administrator adds or selects a department or course.\n3. The administrator enters or updates the record details.\n4. The system validates the fields and saves the record. |
| Alternative Paths | The system rejects missing required values and warns if a record is referenced by other data and cannot safely be deleted. |
| Postcondition | Academic information is available for assignment to relevant users. |
| Exception Paths | Database errors are reported without presenting raw technical details to the user. |
| Other | The exact fields and deletion rules should match the approved database design. |

### 3.2.4 Manage Evaluation Periods

| Field | Specification |
|---|---|
| Requirement ID | 3.2.4 |
| Trigger | The administrator creates or updates an evaluation period. |
| Precondition | The administrator is authenticated and authorized. |
| Basic Path | 1. The system displays evaluation periods.\n2. The administrator enters the school year, semester, opening date, closing date, and status.\n3. The system validates the date range and required fields.\n4. The system saves the period and displays its current status. |

| | |
|---|---|
| Alternative Paths | The system rejects a closing date earlier than the opening date and warns about conflicting active periods if such a rule is configured. |
| Postcondition | The period is saved and its status controls evaluation availability. |
| Exception Paths | If saving fails, the system keeps the existing period data. |
| Other | Opening and closing dates and activation rules require institutional confirmation. |

### 3.2.5 Submit Teacher Evaluation

| Field | Specification |
|---|---|
| Requirement ID | 3.2.5 |
| Trigger | A student opens an available evaluation. |
| Precondition | The student is authenticated; an applicable evaluation period is open; the student is eligible to evaluate the teacher. |
| Basic Path | 1. The system displays the student's eligible evaluation tasks.\n2. The student selects a teacher and the approved evaluation form.\n3. The student completes the required rating items and optional comments, if available.\n4. The student reviews and submits the form.\n5. The system validates answers and checks duplicate-submission rules.\n6. The system stores the evaluation and displays confirmation. |
| Alternative Paths | If required answers are missing, the system identifies the incomplete items. If the period is closed, submission is blocked. |
| Postcondition | The evaluation is saved and marked as submitted. |
| Exception Paths | If storage fails, the system informs the student and does not falsely confirm success. |
| Other | The teacher should not be shown identifying student information in evaluation results unless explicitly approved by CMIC. |

### 3.2.6 View Evaluation Results

| Field | Specification |
|---|---|
| Requirement ID | 3.2.6 |
| Trigger | A teacher opens Evaluation Results, or an administrator requests a report. |
| Precondition | The user is authenticated and has permission to view the requested results. |
| Basic Path | 1. The system identifies the user role and authorized scope.\n2. The system retrieves the available results for the selected teacher and/or period.\n3. The system displays permitted summaries, ratings, and feedback.\n4. The user changes filters or returns to the dashboard as needed. |
| Alternative Paths | If no results are available, the system displays a clear empty-state message. |
| Postcondition | Only authorized results are displayed. |
| Exception Paths | If retrieval fails, the system displays a safe error message and does not expose database details. |
| Other | Release timing, minimum response thresholds, and report export |

| | |
|---|---|
| | rules must be approved by CMIC. |

### 3.2.7 Manage Profile and Settings

| Field | Specification |
|---|---|
| Requirement ID | 3.2.7 |
| Trigger | A user opens their profile or an administrator opens system settings. |
| Precondition | The user is authenticated. |
| Basic Path | 1. The system displays the profile or settings available to the current role.\n2. The user edits permitted fields.\n3. The system validates and saves the changes.\n4. The system displays confirmation. |
| Alternative Paths | The system rejects invalid field values and prevents a user from changing another user's protected account details without authorization. |
| Postcondition | Valid changes are saved. |
| Exception Paths | If an update fails, the system preserves the previous valid values. |
| Other | Profile-picture upload, editable fields, and system-wide settings depend on the final implementation. |

## 3.3 Detailed Non-Functional Requirements

### 3.3.1 Logical Structure of the Data

The following entities represent the principal data groups identified for the project. Field names are indicative and should be aligned with the final database migrations and schema.

| Entity | Example Data Items | Purpose / Notes |
|---|---|---|
| Admin | admin_id, fullname, username, password | Stores authorized administrator accounts. Password value must be securely hashed. |
| Student | student_id, student_no, first_name, last_name, email, username, password, course_id | Stores student account and course association. |
| Teacher | teacher_id, teacher_no, first_name, last_name, email, username, password, department_id | Stores teacher account and department association. |
| Department | department_id, department_name | Stores academic departments. |
| Course | course_id, course_name (and other approved fields) | Stores courses or programs associated with students. |
| Evaluation Period | period_id, school_year, semester, open_date, close_date, status | Defines when evaluations can be submitted. |
| Evaluation Form | form_id and approved form/question fields | Stores the evaluation criteria or questions, depending on final schema. |
| Evaluation | evaluation_id, student_id, teacher_id, period_id, evaluation_at | Records a submitted evaluation and links it to a student, teacher, and period. Access to student linkage must be restricted. |
| Evaluation Details | detail identifier, evaluation/form item reference, rating, comments or response value | Stores individual answers linked to an evaluation and the relevant question/item. |

Relationship overview: a department may be associated with multiple teachers; a course may be associated with multiple students; a student may submit evaluations for multiple teachers across evaluation periods; a teacher may receive multiple evaluations; an evaluation period may contain many evaluation records; each evaluation may contain multiple detail responses. The exact cardinalities and foreign-key names must be verified against the final ER diagram and database schema.

### 3.3.2 Security

- The system shall require authentication before displaying protected pages.
- The system shall enforce role-based authorization on the server side; hiding a navigation link alone is not sufficient protection.
- Passwords shall be stored using an appropriate one-way password hashing mechanism and shall never be displayed in plain text.
- The system shall validate and safely handle user input to reduce risks such as SQL injection and cross-site scripting.
- The system shall use HTTPS in production and avoid exposing secrets or credentials in source code or error messages.
- Evaluation records shall be accessible only to authorized roles. Teacher-facing reports should not identify individual students unless the institution explicitly authorizes this policy.
- The application shall validate uploaded profile images, if upload is enabled, including allowed file types and reasonable file-size limits.
- Logout shall invalidate the authenticated session. Session handling should follow the framework's security practices.
- Administrative changes to accounts, evaluation periods, forms, and results should be restricted to authorized administrators.

### 3.3.3 Usability

- Navigation labels shall be consistent and understandable for the user's role.
- Forms shall use readable text, logical grouping, and clear required-field indicators.
- The system shall provide feedback after successful actions and explain validation errors in plain language.
- The interface should adapt to common desktop and mobile browser sizes where supported.
- The evaluation form should explain its rating scale and submission process before the student submits responses.

### 3.3.4 Reliability and Performance

- The system should save a submitted evaluation only after all required validation checks pass.
- The system should prevent accidental duplicate submissions for the same student, teacher, and period according to the approved policy.
- If the database or network is unavailable, the system should show a clear error and must not display a false success message.
- Under normal expected school usage, common pages and lists should load within a response-time target agreed during testing.
- Backups and recovery procedures should be established by the institution or system administrator before production deployment.

### 3.3.5 Maintainability and Compatibility

- The application should use a consistent naming convention and separate routing, controller, model, and view responsibilities according to the chosen framework.
- Database relationships and validation rules should be documented and maintained with the project source.
- The system should be tested in the browsers and server environment selected for deployment.
- Changes to evaluation questions and periods should be managed by authorized users without requiring changes to unrelated modules, where the design supports configuration.

## Appendix A – Acceptance Criteria

| ID | Acceptance Criterion | Verification Method |
|---|---|---|
| AC-01 | Valid users can sign in and are redirected to the dashboard appropriate to their role. | Test administrator, student, and teacher accounts. |
| AC-02 | Invalid credentials do not grant access to protected pages. | Attempt sign-in with invalid credentials and access protected URLs. |
| AC-03 | Administrator can create and update permitted user and academic records. | Perform create/edit validation tests. |
| AC-04 | Evaluation periods cannot be saved with an invalid date range. | Submit invalid and valid date combinations. |
| AC-05 | Students can submit evaluations only when eligible and the period is open. | Test open, closed, and ineligible scenarios. |
| AC-06 | Required evaluation answers are validated before submission. | Submit an incomplete evaluation. |
| AC-07 | The system confirms successful submission only after saving succeeds. | Test normal and simulated database-failure scenarios. |
| AC-08 | Teachers can view only the results authorized for their account. | Test role permissions and teacher-to-result access. |
| AC-09 | Evaluation results do not reveal student identity contrary to approved CMIC policy. | Inspect teacher-facing screens, reports, and stored/displayed fields. |
| AC-10 | Logout ends the authenticated session. | Log out and attempt to revisit protected pages. |

## Approval and Review

This SRS is a draft baseline for review. The project adviser and authorized CMIC stakeholders should confirm the evaluation workflow, evaluation questions, anonymity policy, report access, duplicate-submission rules, retention policy, and deployment requirements before the document is treated as final.

| Role | Name / Signature | Date |
|---|---|---|
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